Virtual EMG via facial video analysis

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19th International Conference on Image Analysis and Processing
11–15 september 2017
Catania–Italy

EMG: Electromyography

measures the electrical potentials arising from skeletal muscles

Context

Verbal and nonverbal behaviors are used to communicate an internal emotional or affective state.

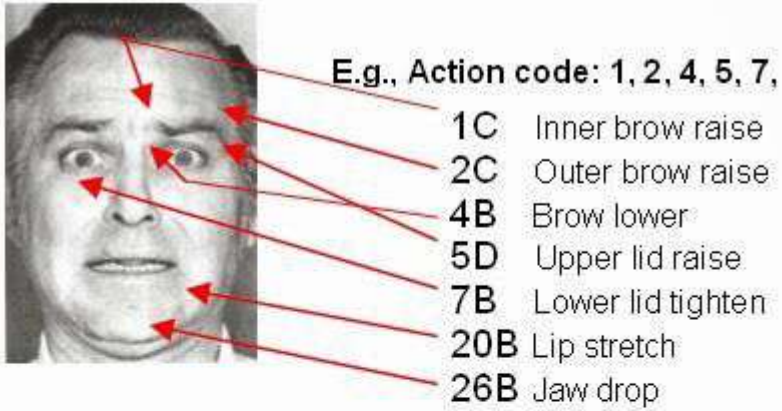
The **face** is the locus of a great deal of emotional expressions.

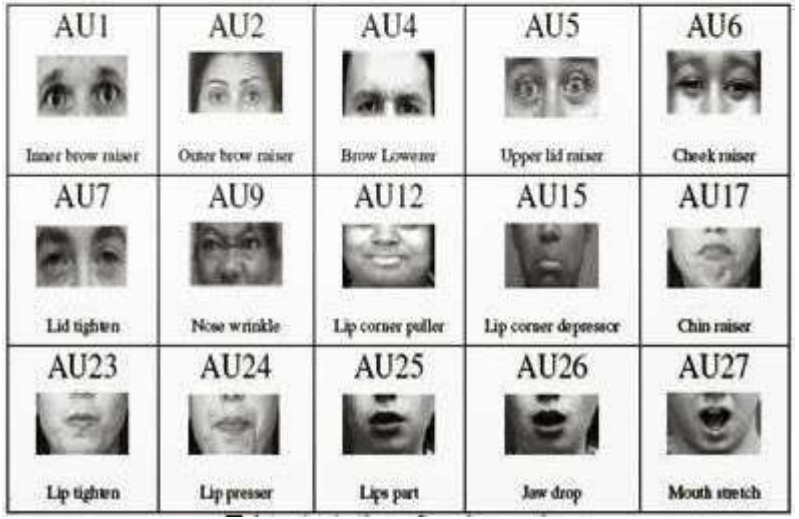
Ekman, P. (1993). Facial expression and emotion. American psychologist, 48(4), 384.

Facial expression

P. Ekman and W.V. Friesen (1978) developed **FACS** (Facial Action Coding System) for describing facial expressions by **action units**.

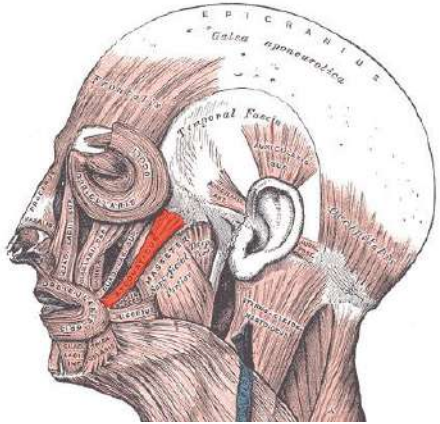
Ekman, P., & Friesen, W. (1978). Facial action coding system: a technique for the measurement of facial movement. Palo Alto: Consulting Psychologists.





Facial muscles

Zygomaticus major muscle

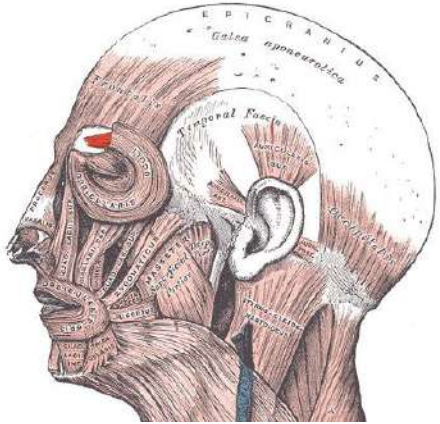


Controls the corners of the mouth and is said to be associated with **positive** emotional valence.

Larsen, J. T., Norris, C. J., & Cacioppo, J. T. (2003). Effects of positive and negative affect on electromyographic activity over zygomaticus major and corrugator supercilii. Psychophysiology, 40(5), 776-785.

Facial muscles

Corrugator supercilii muscle



Lowers the eyebrow and is involved in producing frowns.

Varies **inversely** with the emotional valence.

Larsen, J. T., Norris, C. J., & Cacioppo, J. T. (2003). Effects of positive and negative affect on electromyographic activity over zygomaticus major and corrugator supercilii. Psychophysiology, 40(5), 776-785.

fEMG

Facial EMG (**fEMG**) is a reliable detector of the affective state.

PROS:

- capability of intercepting even very **weak** affective expression
- very good time resolution that allows to register **sudden** expression changes

CONS:

- placing electrodes over the face limits the applicability to **laboratory only**
- is **intrusive** and may alter natural expression

Eskes, M., van Alphen, M. J., Smeele, L. E., Brandsma, D., Balm, A. J., & van der Heijden, F. (2017). Predicting 3D lip movement using facial sEMG: a first step towards estimating functional and aesthetic outcome of oral cancer surgery. Medical & biological engineering & computing, 55(4), 573-583.

Proposed method

Given a video sequence $\mathbf{I}(t)$ of a recorded facial expression and the related set $\mathbf{E}(t)$ of fEMG data:

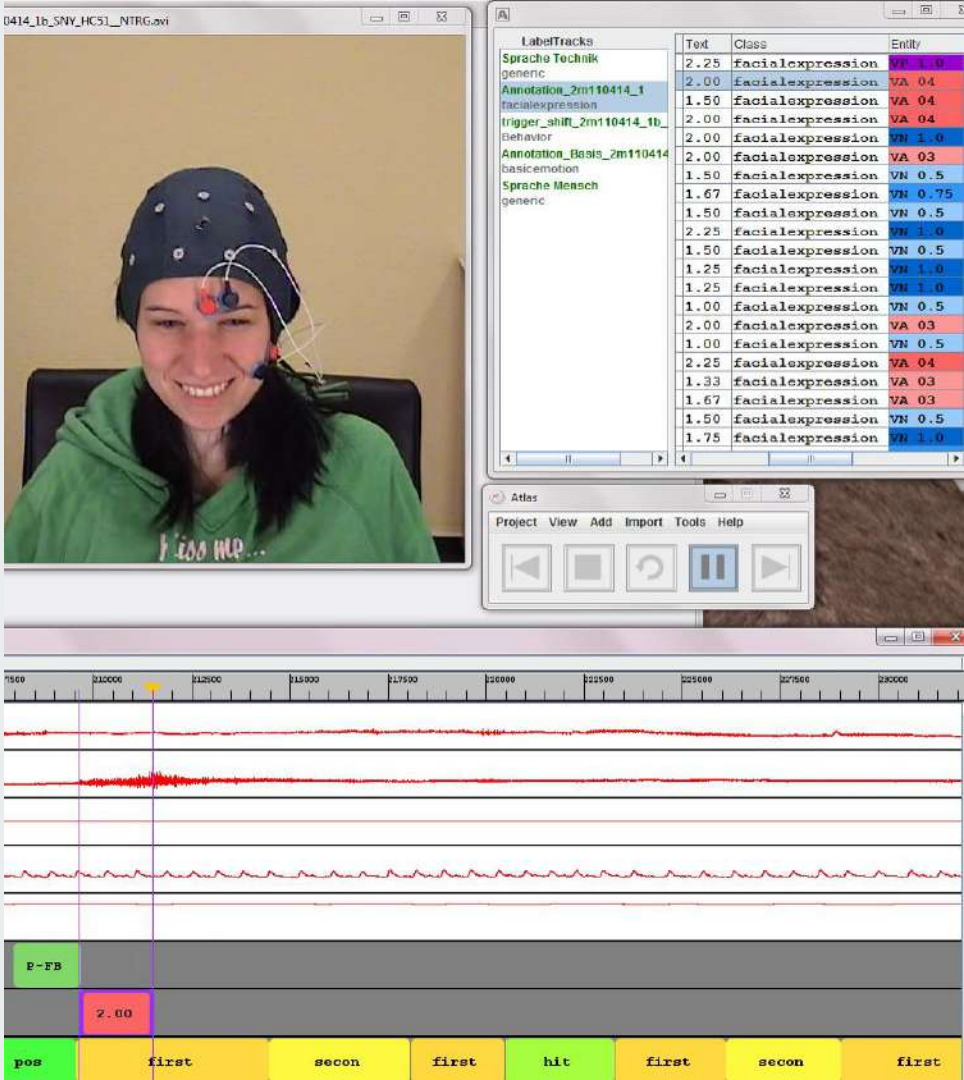
1. Extract from $\mathbf{I}(t)$ the locations of $\mathbf{L}$: a set of facial landmarks
2. Define a pool $\mathbf{l}_p$ of landmarks, on which calculate distance measures $\mathbf{d}_p = \mathbf{d}(\mathbf{l}_p)$
3. Preprocess $\mathbf{E}(t)$ to remove artifacts and downsample to video frequency
4. Train a regression model that maps sequences of landmark distances $\mathbf{D} = \{ \mathbf{d}_{p,s} \mid s = 1 .. S \}$ into fEMG sequences $\mathbf{E} = \{ \mathbf{e}_s \mid s = 1 .. S \}$

Experimental setup

Dataset **OPEN EmoREC II** designed to induce emotional responses in users stimulated by image sequences.

Videos and fEMG signals recorded from **30 subjects**.

Rukavina, S., Gruss, S., Walter, S., Hoffmann, H., & Traue, H. C. (2015). Open_emorec_ii-a multimodal corpus of human-computer interaction. International Journal of Computer, Electrical, Automation, Control and Information Engineering, 9(5), 977-983.



1. Landmark extraction

68 facial landmarks are detected in a **sparse coding** framework, based on Histograms of Sparse Codes (HSC).

$$\mathbf{F}^* \sim P(\mathbf{F} \mid \mathbf{W}, \mathbf{I})$$

$$\mathbf{L}^* = \arg\max_{L} \sum_{k=1}^{m} \int_{\tau} \prod_{i=1}^{l} P(\Delta \mathbf{l}^i_{k,\tau}) P(\mathbf{l}^i | \mathbf{f}^i) d\tau,$$

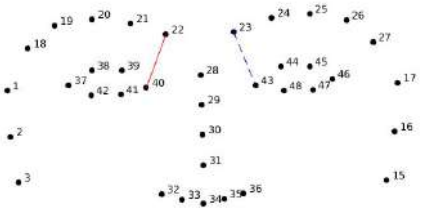
Cuculo, V., Lanzarotti, R., & Boccignone, G. (2014, December). Using sparse coding for landmark localization in facial expressions. In Visual Information Processing (EUVIP), 2014 5th European Workshop on (pp. 1-6). IEEE.

2. Distance measures

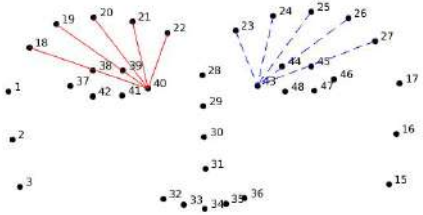
Once the landmarks $\mathbf{L}$ have been detected, pools of landmarks $\mathbf{l}_p$, with $\mathbf{p} \in \{1, \ldots, 6\}$ have been defined for *Corrugator supercilii* and *Zygomaticus major* in order to provide related distance measures $\mathbf{d}_p$ as a "proxy" to muscle activity.

Corrugator supercilii

Pool 1 - 2 - 3 (1+2)

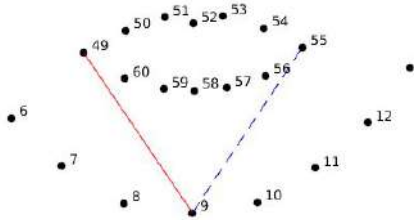


Pool 4 - 5 - 6 (4+5)

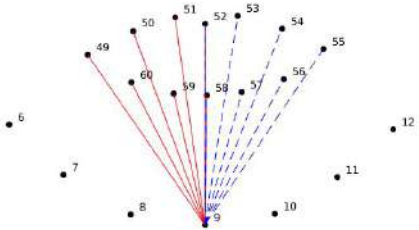


Zygomaticus major

Pool 1 - 2 - 3 (1+2)

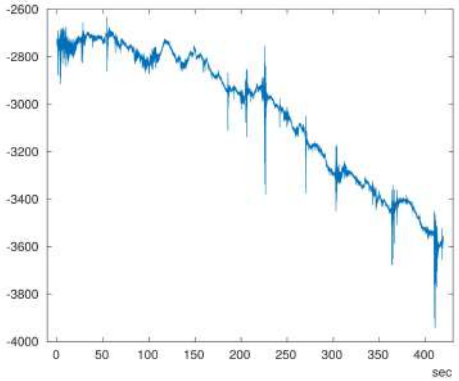


Pool 4 - 5 - 6 (4+5)

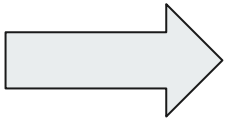


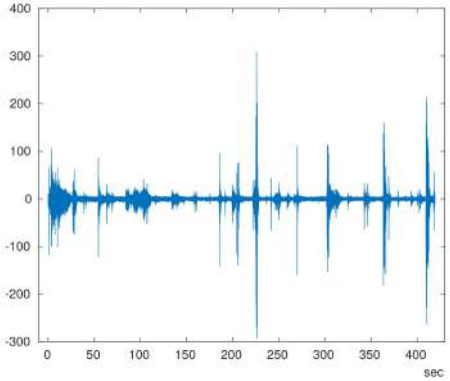
3. fEMG preprocessing

A. 20 Hz high-pass filtering to remove artifacts (eye movements, blinks, swallowing…).
B. 50 Hz notch filter to remove power line interference.



Raw fEMG data

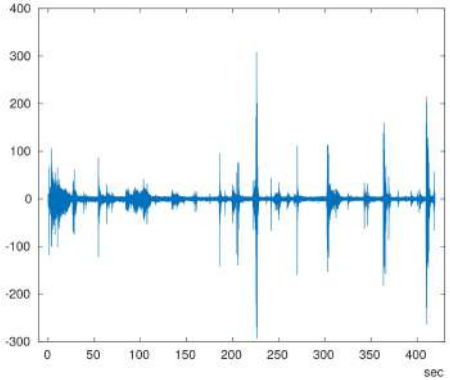




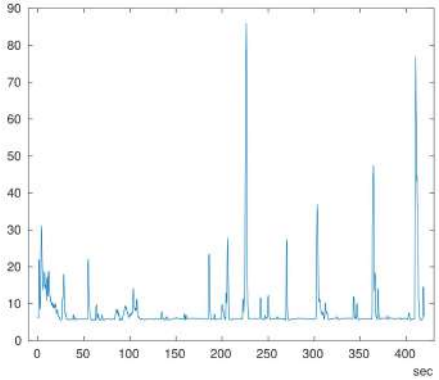
Filtered data

3. fEMG preprocessing

C. Rectification and envelope applied to filtered fEMG.



Filtered data



Enveloped data

D. Downsample to 25 Hz to meet the video frequency.

4. GP model learning and prediction

Given a dataset of $\{\mathbf{L}, \mathbf{E}\} = \{\mathbf{l}_n, \mathbf{e}_n \mid n = 1, \cdots, N\}$, we assume that

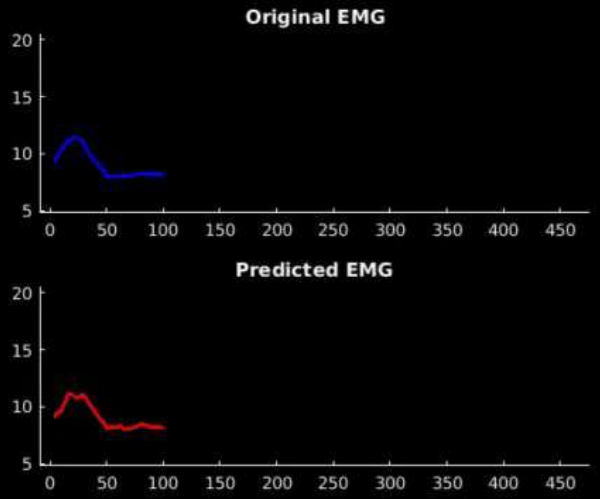
$$e(t) = g(\mathbf{d}(\mathbf{l}_p(t))) + \nu(t), \quad \nu \sim \mathcal{N}(0, \sigma_e^2)$$

and

$$g(\mathbf{d}(\mathbf{l}_p)) \sim \mathcal{GP}(\mu(\mathbf{d}(\mathbf{l}_p)), k(\mathbf{d}(\mathbf{l}_p), \mathbf{d}'(\mathbf{l}_p))),$$

with kernel functions and related hyperparameters expressed in closed form and obtained from the training stage.

Rasmussen, C. E., & Williams, C. K. (2006). Gaussian processes for machine learning (Vol. 1). Cambridge: MIT press.


Original EMG
20
15
10
5
0
50
100
150
200
250
300
350
400
450
Predicted EMG
20
15
10
5
0
50
100
150
200
250
300
350
400
450

Results

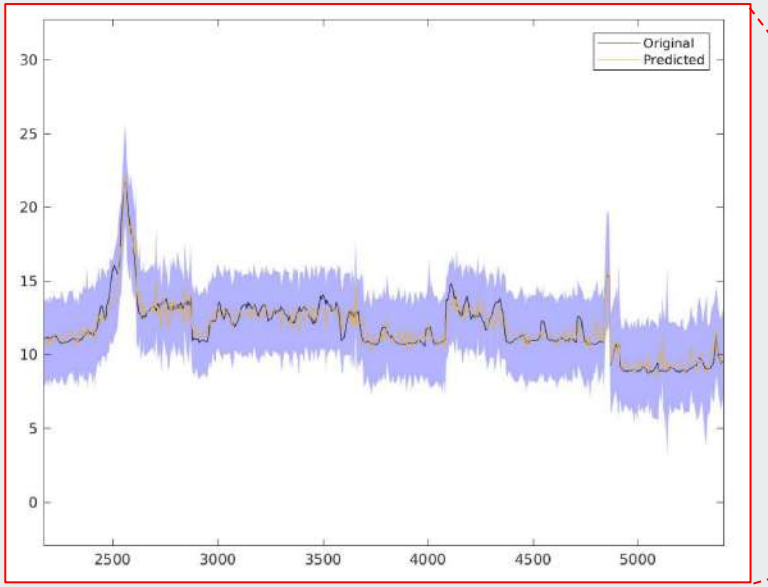
Results evaluated in terms of Mean Square Error (MSE) and Concordance Correlation Coefficient (CCC)

$$MSE(\mathbf{e}, \widetilde{\mathbf{e}}) = \frac{1}{T}\sum_{t=1}^{T}(e(t) - \tilde{e}(t))^2 \qquad CCC(\mathbf{e}, \widetilde{\mathbf{e}}) = \frac{2cov(\mathbf{e}, \widetilde{\mathbf{e}})}{\sigma_{\mathbf{e}}^2 + \sigma_{\tilde{\mathbf{e}}}^2 + (\mu_{\mathbf{e}} - \mu_{\tilde{\mathbf{e}}})^2},$$

training different models for each of the **6 pools,** each of **3 different kernel functions** and **2 considered muscles**.

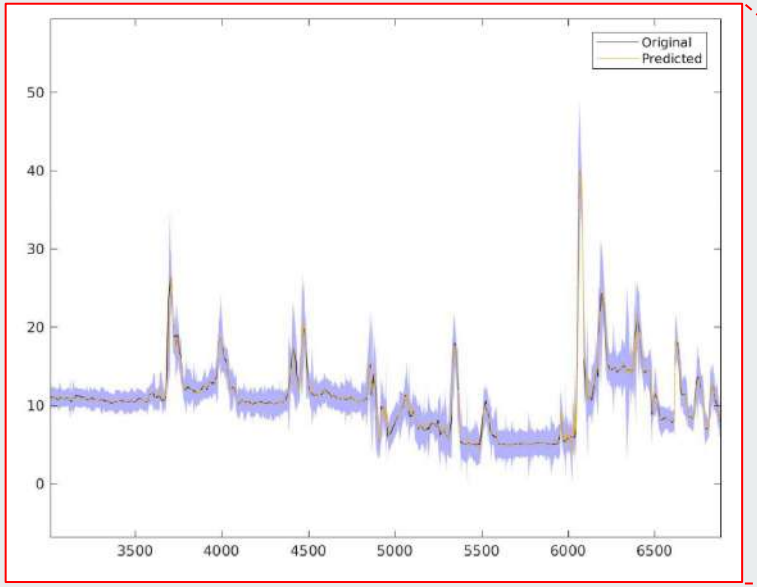
Partitioning data with k-fold cross validation, *k*=10.

Corrugator supercilii



P	MSE			CCC		
	SE	RQ	M32	SE	RQ	M32
1	42.007	31.806	36.226	0.9855	0.9888	0.9873
2	4.962	4.928	4.934	0.9983	0.9983	0.9983
3	3.828	3.760	3.718	0.9987	0.9987	0.9987
4	2.008	1.769	1.727	0.9993	0.9994	0.9994
5	1.968	1.799	1.707	0.9993	0.9994	0.9994
6	1.047	0.716	**0.652**	0.9996	0.9997	**0.9998**

Zygomaticus major



P	MSE			CCC		
	SE	RQ	M32	SE	RQ	M32
1	11.540	11.519	11.500	0.9969	0.9969	0.9969
2	13.331	13.226	13.192	0.9965	0.9965	0.9965
3	8.197	7.825	7.787	0.9978	0.9979	0.9979
4	2.254	1.572	1.421	0.9994	0.9996	0.9996
5	2.540	1.276	1.319	0.9993	0.9997	0.9997
6	1.040	0.629	**0.561**	0.9997	0.9998	**0.9999**

Conclusions

- This is the **first attempt*** to estimate fEMG signals from video sequences.
- Monitoring physiological signals via **non contact** means has promise for a variety of out-of-lab applications.
- Real fEMG can intercept even **very weak** affective expressions.
- Usage of **depth** cameras or **high-speed** cameras could give better results.

*to the best of our knowledge

Thanks

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